

## EXPERIMENT 1 TO 3 – MATHIMOZHI ER

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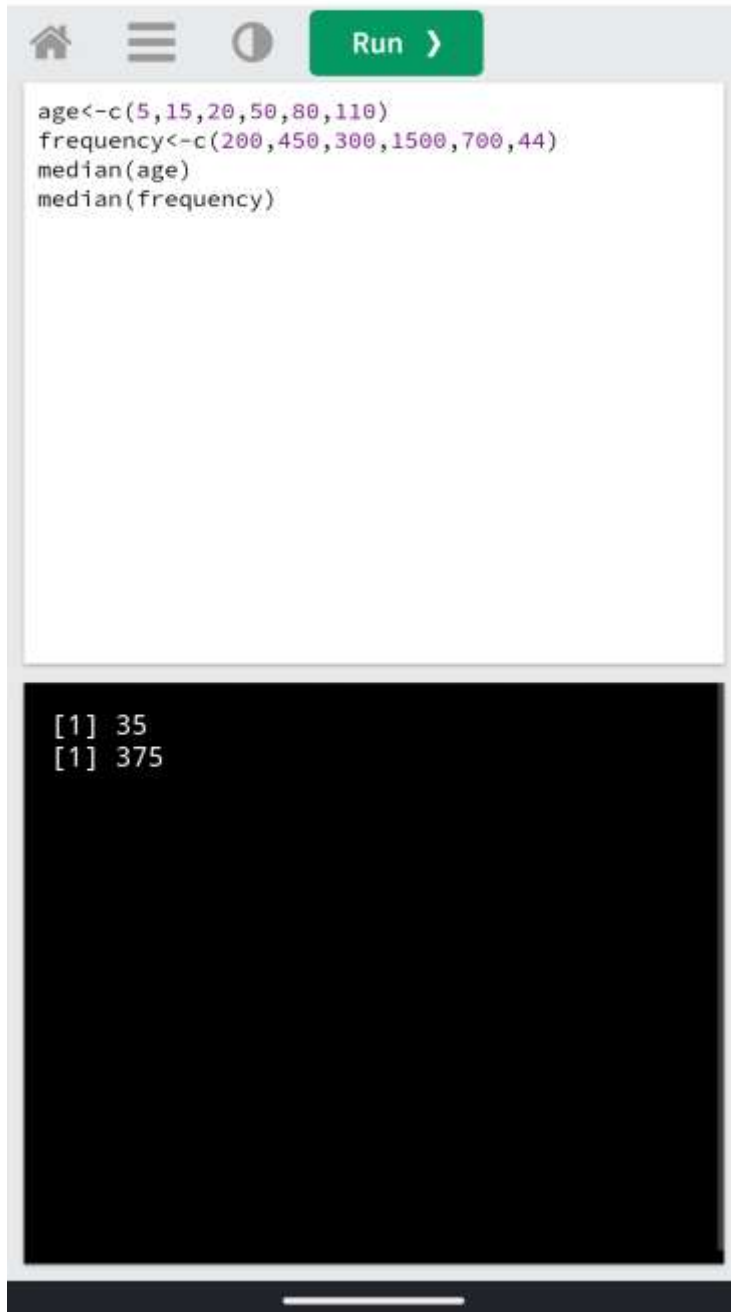
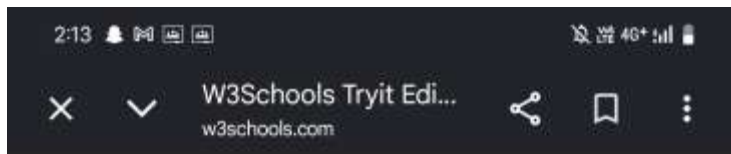
### Experiment-1

**AIM:** To find intervals and corresponding frequencies.

#### Procedure:

1. Collect and Sort Data: Collect the raw data and arrange it in ascending order to find the minimum and maximum value.
2. Find Range and Number of Classes: Calculate Range = Max - Min. Decide number of classes (k) using Sturges' formula  $k = 1 + 3.322 \sqrt{\log N}$ , where N is total observations.
3. Calculate Class Interval Size: Find class width  $h = \text{Range} / k$  and round it to a convenient number. Then form the continuous class intervals like 0-10, 10-20, etc.
4. Prepare Frequency Distribution: Use tally marks method to count how many values fall into each class interval and note the corresponding frequency.
5. Verify and Tabulate: Check that sum of all frequencies equals total N. Finally prepare a frequency distribution table with Class Interval and Frequency columns.

#### Program and Output:



**Result:**

It is executed successfully.

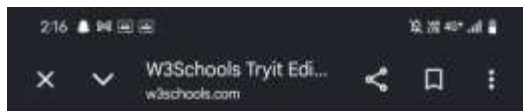
## EXPERIMENT – 2:

AIM: To find the data for analysis includes the attribute age.

Procedure:

1. Data Collection: Collect the raw dataset from the given source (CSV / Database / Survey) and check if it contains the attribute Age.
2. Data Extraction: Filter and extract only the Age column along with relevant identifier columns (like ID, Name) for analysis purpose.
3. Data Cleaning: Remove missing values, null values and invalid ages (e.g., negative values or age > 100) from the age attribute to get clean data.
4. Data Grouping: Group the age data into meaningful intervals like 0-18, 19-35, 36-60, 60+ to make analysis easier and count frequency for each group.
5. Analysis and Interpretation: Calculate mean, median, mode of age and plot histogram/bar chart to analyze the age distribution pattern for decision making.

**PROGRAM & OUTPUT:**



```
age<-  
c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,  
33,33,35,35,35,35,36,48,45,46,52,70)  
mean(age)  
median(age)  
mode_age<-names(table(age))  
[table(age)==max(table(age))]  
mode_age  
range(age)  
quantile(age,.25)  
quantile(age,.75)
```

```
[1] 29.96296  
[1] 25  
[1] "25" "35"  
[1] 13 70  
25%  
20.5  
75%  
35
```

**RESULT:** It is executed successfully.

### Experiment-3

**AIM:-** To show Data Preprocessing in Reduction and Transformation.

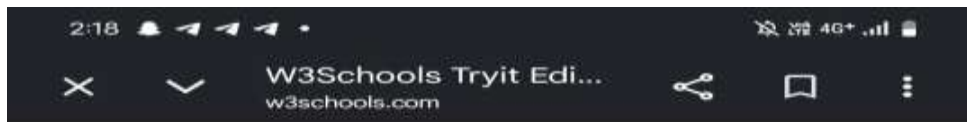
**Objective:** To perform data smoothing by binning method using mean, median and min-max to reduce noise.

#### Procedure:

1. Take raw numeric data which contains noisy values.

2. Apply Equal-Width Binning by dividing data into 5 bins using cut() function.
3. Perform smoothing by bin means - replace each value in bin by mean of that bin.
4. Perform smoothing by bin median - replace each value by median of bin to handle outliers.
5. Perform smoothing by bin boundaries - replace each value by nearest min/max of its bin.

**PROGRAM & OUTPUT:**

A screenshot of a code editor interface. At the top, there is a home icon, a menu icon, a moon icon, and a green 'Run' button with a right arrow. Below the toolbar is a text area containing R code. The code defines a vector 'data', calculates the number of bins, and then applies mean, median, and min-max smoothing functions to the data using 'tapply'. Below the code area is a black terminal window showing the output of the code execution. The output displays the smoothed values for each bin and the corresponding index ranges.

```
data <-
c(11,13,13,15,15,16,19,20,20,20,21,21,22,23,24,
30,40,45,45,45,71,72,73,75)
bins <- 5
bin_indices <- cut(data, bins)
mean_smooth <- tapply(data, bin_indices, mean)
print(mean_smooth)
median_smooth <- tapply(data, bin_indices,
median)
median_smooth
min_max_smooth <- tapply(data, bin_indices,
function(x) c(min(x), max(x)))
print(min_max_smooth)
```

```
(10.9,23.8] (23.8,36.6] (36.6,49.4] (49.4
17.78571 27.00000 43.75000
(10.9,23.8] (23.8,36.6] (36.6,49.4] (49.4
19.5 27.0 45.0
$`(10.9,23.8]`
[1] 11 23

$`(23.8,36.6]`
[1] 24 30

$`(36.6,49.4]`
[1] 40 45

$`(49.4,62.2]`
NULL

$`(62.2,75.1]`
```

**RESULT:** The program for Data Preprocessing using Data Reduction and Transformation by Binning method has been executed successfully and noise is reduced by smoothing.